VHF-FM System (S.O.O. 6296) – General data

1. General

The Very High Frequency Modulation (VHF-FM) system is used for voice communication between aircraft and/or aircraft and a ground station in the 138.000 to 174.000 MHz frequency range of the VHF band.

2. Description

A. VHF-FM

The VHF-FM transceiver provides secondary airborne voice communications on crew selectable frequencies, operating in the 138.000 MHz to 174.000 MHz frequency range in 2.5 kHz increments, with either 25 kHz or 12.5 kHz channel spacing. The transmitter section of this unit has a minimum of 8 watts and does not exceed 10 watts output power. Output power may be reduced by a front panel switch to 1 watt, in order to reduce interference to land based systems.

The VHF-FM system includes a self-contained transceiver installed in the cockpit instrument panel, a blade style antenna installed on the cabin roof near the LH wing root, and a circuit breaker (3 amp) labeled 'VHF-FM' installed in the avionics circuit breaker panel.

The VHF-FM transceiver provides 120 operator accessible memory positions, each of which is capable of storing a transmit frequency, receive frequency, transmit frequency CTCSS (Continuous Tone-Coded Squelch System) tone or DPL (Digital Private Line) code, receive frequency CTCSS tone or DPL code, an alphanumeric identifier for each channel and in the TFM–138B, wideband (25 kHz) or narrowband (12.5 kHz) channel spacing assignment. Operating frequency and other related data are presented on a 48 character, two line LED matrix display.

Data entry and function control are done via a 12 button keypad on the front panel. Present channels may also be scrolled and scanned using the keyboard. Data may also be entered via a PC computer with the provided software and optional PIB–100 programming interface box.

3. Adjustment/Test

A. VHF-FM

1) Special Tools

- VHF-FM Ramp Test Set with antenna. (IFR AM/FM–1000, Aeroflex IFR–4000 or equivalent)
- In Line Wattmeter (Bird 43 Wattmeter with 10C or 25C element)

2) Function Test

a) Apply DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

b) Wait several minutes for the aircraft electrical and avionics systems to initialize.

c) Ensure that VHF-FM (Q6) (Right Bus) circuit breaker on the avionics circuit breaker panel is closed.

d) Set up VHF-FM Communication ramp test unit, in accordance with the instructions in the test set manual.

e) Tune the VHF-FM Communication ramp test unit to an unused frequency.

f) Tune the VHF-FM transceiver channel to match the ramp test unit frequency. Adjust volume and squelch to a comfortable level.

g) Enable a MOD TONE on the ramp test unit. Verify a tone is heard when the appropriate COM channel is selected on the audio control panel.

h) Do a VSWR check on the VHF-FM antenna using an inline type Wattmeter. The check should be performed with the LO/HI power switch in the HI position. Reflected power should not exceed 10% of the forward power.

i) Tune the VHF-FM transceiver to a local frequency and do a radio maintenance check. Transmit audio, receive audio and sidetone audio should be clear, undistorted, and at a comfortable level.

j) Disconnect DC power (refer to 12-00-00, Servicing – General Data, Para 2.).

B. Electrical Bonding Test – VHF-FM Antenna

1) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

2) Gain access to the ceiling compartment aft of station 219.5. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

3) Verify that the electrical conductivity is consistent over the electrical bond contact area by placing a probe on the VHF antenna mounting screw. Place the second probe on the nearest ground stud. The resistance between the two points must be <3.0 milliohms unless stated otherwise.

4) Check the electrical conductivity between several random mounting screws within the electrical bond contact area to ensure consistency.

5) If the electrical conductivity is not consistent over the electrical bond contact area, check the antenna electrical bonding in accordance with 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General Data.

6) Replace ceiling compartment. (Refer to 25-20-00, Cabin Compartment – Commuter Version (Mod 6/2050) – General Data).

4. Removal/Installation

A. VHF-FM Antenna (CI–295–200)

Refer to Figure 1 for details.

1) Removal

a) Ensure the aircraft system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Open VHF-FM (Q6) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clip.

c) Carefully cut the sealant from around the base of the antenna.

d) Unscrew mounting screws (6) from antenna assembly.

e) Carefully lift the antenna clear of the fuselage until there is access to the coaxial connection.

f) Disconnect the coaxial connector.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure VHF-FM (Q6) (Right Bus) circuit breaker on the avionics circuit breaker panel is opened and fitted with a safety clip.

c) Solvent clean the electrical contact bond areas immediately before the installation. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

d) Move the antenna assembly close to the fuselage, connect the coaxial connector.

e) Secure the antenna assembly on fuselage with the mounting screws (6).

f) Solvent clean the exposed area of the airframe around the antenna. (Refer to 20-75-00, Electrical Bonding and Grounding of Aircraft Structures – General data).

g) If required, touch up the exposed area of airframe around the antenna with F19 primers by brush within 24 hours of the antenna installation. Ensure that the primer re-finishing provides complete coverage of the affected area.

h) If the affected area requires paint touch up, brush coat the area with the top-coat paint system matching the existing.

i) Seal around the antenna base and dome seal the securing screws. (Refer to 20-80-00, Installation of Aircraft Antennas – General data).

j) Check the antenna electrical bonding as per Para 3.B.

k) Remove the safety clips and close VHF-FM (Q6) (Right Bus) circuit breaker.

l) Do a function test on the VHF-FM system as per Para 3.A.2).

B. VHF-FM Transceiver (TFM–138B)

1) Removal

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Pull the VHF-FM (Q6) (Right Bus) circuit breaker on the avionics circuit breaker panel and fit safety clips.

c) Using an appropriate tool, push and turn each of the DZUS fasteners securing the unit counterclockwise until the unit disengages from the mounting.

d) Remove the transceiver unit from the mounting.

e) Remove unit from aircraft.

Note Stow the connectors of the removed components.

2) Installation

a) Ensure the aircraft electrical system is OFF. (Refer to 24-00-00, Electrical Power – General Data).

b) Ensure that the VHF-FM (Q6) (Right Bus) circuit breaker on the avionics circuit breaker panel is pulled and fitted with safety clips.

c) Insert the TFM–138B unit into its mounting in the instrument panel.

d) Using an appropriate tool, push and turn each of the DZUS fasteners clockwise until the unit is secure in the mounting.

e) Remove safety clips and close VHF-FM (Q6) (Right Bus) circuit breaker on the avionics circuit breaker panel.

f) Do a function test on the VHF-FM system as per Para 3.A.2).

5. Troubleshooting

A. VHF-FM

1) If the VHF-FM Transceiver (TFM–138B) is deemed inoperative, check the following:

a) Ensure the VHF-FM (Q6) circuit breaker is closed.

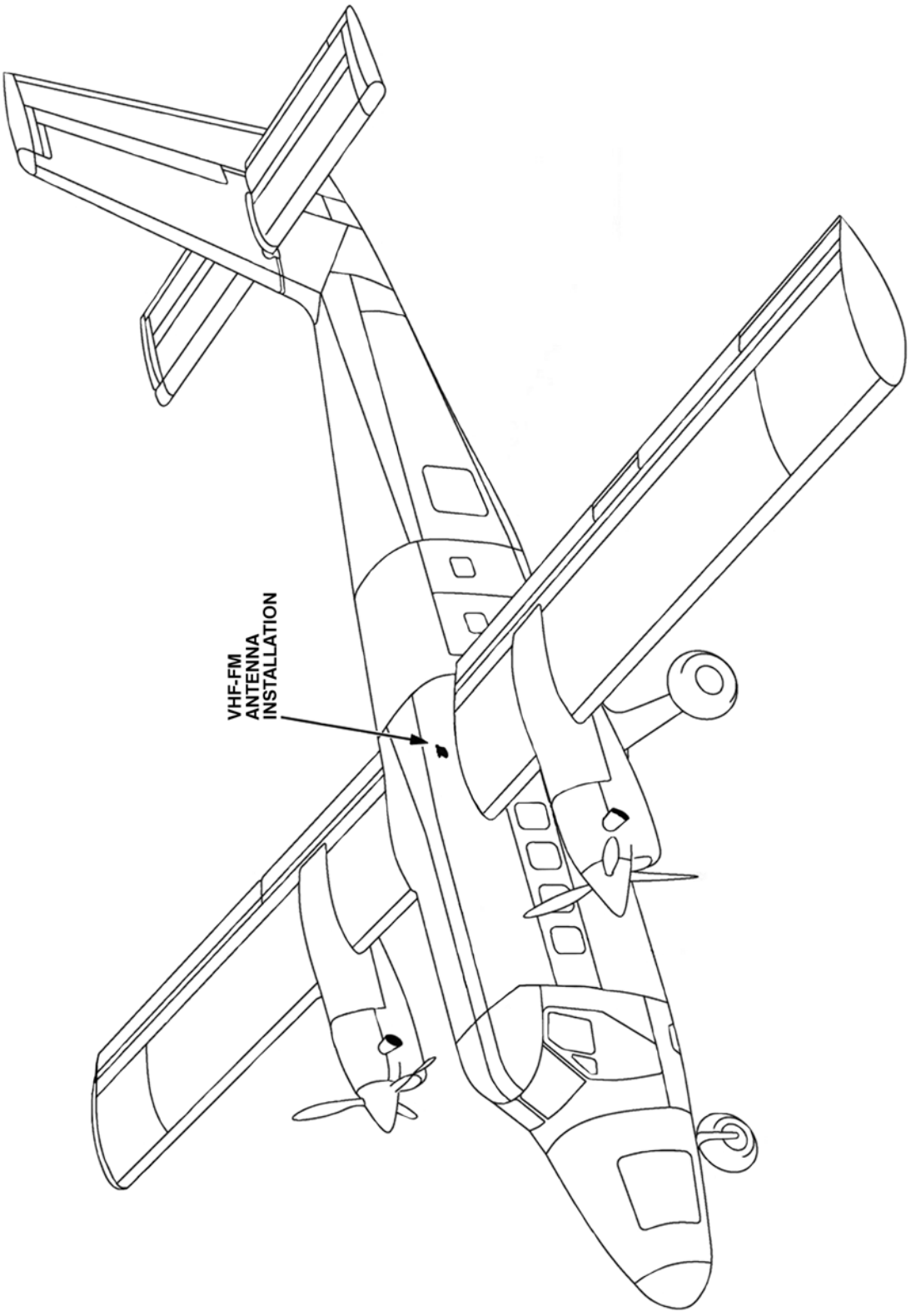
b) Visually inspect the VHF-FM unit for damaged wires or contacts.

c) Ensure that the VHF-FM connectors are properly mated and secure.

d) Ensure antenna connections are properly mated and secure.

e) If troubleshooting at the wire level is required, refer to the applicable wiring diagrams in Wiring Manual, PSM 1-64-2W.

f) If repair or service is required, send to appropriate rated avionics repair facility.



VHF-FM Antenna Location (S.O.O. 6296)
Figure 1

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